

Groups (2015–2021)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

51

TOTAL MARKS

82 M

TEST DURATION

153 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	23	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	9	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	19	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q51 • ID: JAM_2021_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 The number of elements of order two in the group S_4 is equal to ____.

Question 2 (JAM 2021 • Q47 • ID: JAM_2021_Q47)

NAT

+1 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 47 The number of group homomorphisms from the group $\mathbb{Z}_4$ to the group S_3 is ____.

Question 3 (JAM 2021 • Q41 • ID: JAM_2021_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 The number of cycles of length 4 in S_6 is ____.

Question 4 (JAM 2021 • Q36 • ID: JAM_2021_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 Let G be a finite group of order 28. Assume that G contains a subgroup of order 7. Which of the following statements is/are true?

- (A) G contains a unique subgroup of order 7.
- (B) G contains a normal subgroup of order 7.
- (C) G contains no normal subgroup of order 7.
- (D) G contains at least two subgroups of order 7.

Question 5 (JAM 2021 • Q28 • ID: JAM_2021_Q28)

MCQ

+2 M

Neg: -0.67

Q. 28 Let G be a finite abelian group of odd order. Consider the following two statements:

I. The map $f : G \rightarrow G$ defined by $f(g) = g^2$ is a group isomorphism.

II. The product $\prod_{g \in G} g = e$.

(A) Both I and II are TRUE.

(B) I is TRUE but II is FALSE.

(C) II is TRUE but I is FALSE.

(D) Neither I nor II is TRUE.

MA

9 / 17

Question 6 (JAM 2021 • Q27 • ID: JAM_2021_Q27)

MCQ

+2 M

Neg: -0.67

Q. 27 Let g be an element of S_7 such that g commutes with the element $(2, 6, 4, 3)$. The number of such g is

(A) 6.

(B) 4.

(C) 24.

(D) 48.

Question 7 (JAM 2021 • Q24 • ID: JAM_2021_Q24)

MCQ

+2 M

Neg: -0.67

Q. 24 Which one of the following statements is true?

(A) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{R}, +)$.

(B) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.

(C) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}/2\mathbb{Z}, +)$.

(D) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.

MA

8 / 17

Question 8 (JAM 2021 • Q20 • ID: JAM_2021_Q20)

MCQ

+2 M

Neg: -0.67

Q. 20 Which one of the following statements is true?

- (A) Exactly half of the elements in any even order subgroup of S_5 must be even permutations.
- (B) Any abelian subgroup of S_5 is trivial.
- (C) There exists a cyclic subgroup of S_5 of order 6.
- (D) There exists a normal subgroup of S_5 of index 7.

Question 9 (JAM 2021 • Q14 • ID: JAM_2021_Q14)

MCQ

+2 M

Neg: -0.67

Q. 14 Consider the following statements.

- I. The group $(\mathbb{Q}, +)$ has no proper subgroup of finite index.
- II. The group $(\mathbb{C} \setminus \{0\}, \cdot)$ has no proper subgroup of finite index.

Which one of the following statements is true?

- (A) Both I and II are TRUE.
- (B) I is TRUE but II is FALSE.
- (C) II is TRUE but I is FALSE.
- (D) Neither I nor II is TRUE.

MA

5 / 17

Question 10 (JAM 2021 • Q6 • ID: JAM_2021_Q6)

MCQ

+1 M

Neg: -0.33

Q. 6 How many elements of the group $\mathbb{Z}_{50}$ have order 10?

(A) 10

(B) 4

(C) 5

(D) 8



MA

2 / 17

Question 11 (JAM 2020 • Q60 • ID: JAM_2020_Q60)

NAT

+2 M

Neg: Nil (0)

Q. 60 Suppose that G is a group of order 57 which is NOT cyclic. If G contains a unique subgroup H of order 19, then for any $g \notin H$, $o(g)$ is _____.

END OF THE QUESTION PAPER

MA

17 / 17

Question 12 (JAM 2020 • Q50 • ID: JAM_2020_Q50)

NAT

+1 M

Neg: Nil (0)

Q. 50 Let $\phi : S_3 \rightarrow S^1$ be a non-trivial non-injective group homomorphism. Then, the number of elements in the kernel of ϕ is _____.

MA

15 / 17

Question 13 (JAM 2020 • Q40 • ID: JAM_2020_Q40)

MSQ

+2 M

Neg: Nil (0)

Q. 40 Let G be a group with identity e . Let H be an abelian non-trivial proper subgroup of G with the property that $H \cap gHg^{-1} = \{e\}$ for all $g \notin H$.

If $K = \{g \in G : gh = hg \text{ for all } h \in H\}$, then

(A) K is a proper subgroup of H

(B) H is a proper subgroup of K

(C) $K = H$

(D) there exists no abelian subgroup $L \subseteq G$ such that K is a proper subgroup of L

MA

13 / 17

Question 14 (JAM 2020 • Q30 • ID: JAM_2020_Q30)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 30 Let $F = \{\omega \in \mathbb{C} : \omega^{2020} = 1\}$. Consider the groups

$$G = \left\{ \begin{pmatrix} \omega & z \\ 0 & 1 \end{pmatrix} : \omega \in F, z \in \mathbb{C} \right\}$$

and

$$H = \left\{ \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} : z \in \mathbb{C} \right\}$$

under matrix multiplication. Then the number of cosets of H in G is

(A) 1010

(B) 2019

(C) 2020

(D) infinite

MA

10 / 17

Question 15 (JAM 2020 • Q29 • ID: JAM_2020_Q29)

MCQ

+2 M

Neg: -0.67

Q. 29 Let $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ be the circle group under multiplication and $i = \sqrt{-1}$. Then the set $\{\theta \in \mathbb{R} : \langle e^{i2\pi\theta} \rangle \text{ is infinite}\}$ is

(A) empty

(B) non-empty and finite

(C) countably infinite

(D) uncountable

MA

9 / 17

Question 16 (JAM 2020 • Q17 • ID: JAM_2020_Q17)

MCQ

+2 M

Neg: -0.67

Q. 17 Let $f(x, y, z) = x^3 + y^3 + z^3 - 3xyz$. A point at which the gradient of the function f is equal to zero is

(A) $(-1, 1, -1)$ (B) $(-1, -1, -1)$ (C) $(-1, 1, 1)$ (D) $(1, -1, 1)$

Question 17 (JAM 2020 • Q10 • ID: JAM_2020_Q10)

MCQ

+1 M

Neg: -0.33

JAM 2020

MATHEMATICS - MA

Q. 10 Consider the following group under matrix multiplication:

$$H = \left\{ \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix} : p, q, r \in \mathbb{R} \right\}.$$

Then the center of the group is isomorphic to

(A) $(\mathbb{R} \setminus \{0\}, \times)$ (B) $(\mathbb{R}, +)$ (C) $(\mathbb{R}^2, +)$ (D) $(\mathbb{R}, +) \times (\mathbb{R} \setminus \{0\}, \times)$

Q. 11 – Q. 30 carry two marks each.

Question 18 (JAM 2020 • Q8 • ID: JAM_2020_Q8)

MCQ

+1 M

Neg: -0.33

Q. 8 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation given by $T(x, y) = (-x, y)$. Then(A) $T^{2k} = T$ for all $k \geq 1$ (B) $T^{2k+1} = -T$ for all $k \geq 1$ (C) the range of T^2 is a proper subspace of the range of T (D) the range of T^2 is equal to the range of T

Question 19 (JAM 2019 • Q55 • ID: JAM_2019_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55 The number of elements in the set $\{x \in S_3 : x^4 = e\}$, where e is the identity element of the permutation group S_3 , is _____

Question 20 (JAM 2019 • Q53 • ID: JAM_2019_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let $G = \{n \in \mathbb{N} : n \leq 55, \gcd(n, 55) = 1\}$ be the group under multiplication modulo 55. Let $x \in G$ be such that $x^2 = 26$ and $x > 30$. Then x is equal to _____

Question 21 (JAM 2019 • Q41 • ID: JAM_2019_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 Let x be the 100-cycle $(1\ 2\ 3\ \cdots\ 100)$ and let y be the transposition $(49\ 50)$ in the permutation group S_{100} . Then the order of xy is _____

Question 22 (JAM 2019 • Q32 • ID: JAM_2019_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 Let G be a nonabelian group, $y \in G$, and let the maps f, g, h from G to itself be defined by

$$f(x) = yxy^{-1}, \quad g(x) = x^{-1} \quad \text{and} \quad h = g \circ g.$$

Then

- (A) g and h are homomorphisms and f is not a homomorphism
- (B) h is a homomorphism and g is not a homomorphism
- (C) f is a homomorphism and g is not a homomorphism
- (D) f, g and h are homomorphisms

Question 23 (JAM 2019 • Q25 • ID: JAM_2019_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 Let $\vec{F}(x, y, z) = 2y\hat{i} + x^2\hat{j} + xy\hat{k}$ and let $\mathcal{C}$ be the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 1$. Then the value of

$$\left| \oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} \right|$$

is

- (A) π
- (B) $\frac{3\pi}{2}$
- (C) 2π
- (D) 3π

Question 24 (JAM 2019 • Q14 • ID: JAM_2019_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 The set $\left\{\frac{1}{m} + \frac{1}{n} : m, n \in \mathbb{N}\right\} \cup \{0\}$, as a subset of $\mathbb{R}$, is

- (A) compact and open (B) compact but not open
(C) not compact but open (D) neither compact nor open

Question 25 (JAM 2019 • Q3 • ID: JAM_2019_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. Define $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$f(x, y, z) = g(x^2 + y^2 - 2z^2).$$

Then $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ is equal to

- (A) $4(x^2 + y^2 - 4z^2) g''(x^2 + y^2 - 2z^2)$
(B) $4(x^2 + y^2 + 4z^2) g''(x^2 + y^2 - 2z^2)$
(C) $4(x^2 + y^2 - 2z^2) g''(x^2 + y^2 - 2z^2)$
(D) $4(x^2 + y^2 + 4z^2) g''(x^2 + y^2 - 2z^2) + 8g'(x^2 + y^2 - 2z^2)$

Question 26 (JAM 2018 • Q48 • ID: JAM_2018_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let A_6 be the group of even permutations of 6 distinct symbols. Then the number of elements of order 6 in A_6 is _____

Question 27 (JAM 2018 • Q41 • ID: JAM_2018_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 The order of the element $(1\ 2\ 3)(2\ 4\ 5)(4\ 5\ 6)$ in the group S_6 is _____

Question 28 (JAM 2018 • Q35 • ID: JAM_2018_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35 Let $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ denote the group of non-zero complex numbers under multiplication. Suppose

$Y_n = \{z \in \mathbb{C} \mid z^n = 1\}$, $n \in \mathbb{N}$. Which of the following is (are) subgroup(s) of $\mathbb{C}^*$?

- (A) $\bigcup_{n=1}^{100} Y_n$ (B) $\bigcup_{n=1}^{\infty} Y_{2^n}$ (C) $\bigcup_{n=100}^{\infty} Y_n$ (D) $\bigcup_{n=1}^{\infty} Y_n$

Question 29 (JAM 2018 • Q33 • ID: JAM_2018_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Suppose f, g, h are permutations of the set $\{\alpha, \beta, \gamma, \delta\}$, where

f interchanges α and β but fixes γ and δ ,

g interchanges β and γ but fixes α and δ ,

h interchanges γ and δ but fixes α and β .

Which of the following permutations interchange(s) α and δ but fix(es) β and γ ?

- (A) $f \circ g \circ h \circ g \circ f$ (B) $g \circ h \circ f \circ h \circ g$ (C) $g \circ f \circ h \circ f \circ g$ (D) $h \circ g \circ f \circ g \circ h$

Question 30 (JAM 2018 • Q25 • ID: JAM_2018_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 Let G be a group satisfying the property that $f: G \rightarrow \mathbb{Z}_{221}$ is a homomorphism implies

$f(g) = 0$, $\forall g \in G$. Then a possible group G is

- (A) $\mathbb{Z}_{21}$ (B) $\mathbb{Z}_{51}$ (C) $\mathbb{Z}_{91}$ (D) $\mathbb{Z}_{119}$

MA

6/12

Question 31 (JAM 2018 • Q15 • ID: JAM_2018_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 Suppose that $f, g: \mathbb{R} \rightarrow \mathbb{R}$ are differentiable functions such that f is strictly increasing and g is strictly decreasing. Define $p(x) = f(g(x))$ and $q(x) = g(f(x))$, $\forall x \in \mathbb{R}$. Then, for $t > 0$, the sign of $\int_0^t p'(x) (q'(x) - 3) dx$ is

- (A) positive (B) negative (C) dependent on t (D) dependent on f and g

Question 32 (JAM 2018 • Q5 • ID: JAM_2018_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 The tangent plane to the surface $z = \sqrt{x^2 + 3y^2}$ at $(1, 1, 2)$ is given by

(A) $x - 3y + z = 0$

(B) $x + 3y - 2z = 0$

(C) $2x + 4y - 3z = 0$

(D) $3x - 7y + 2z = 0$

MA

2/12

Question 33 (JAM 2017 • Q54 • ID: JAM_2017_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 The maximum order of a permutation σ in the symmetric group S_{10} is _____

Question 34 (JAM 2017 • Q50 • ID: JAM_2017_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 The number of subgroups of $\mathbb{Z}_7 \times \mathbb{Z}_7$ of order 7 is _____

Question 35 (JAM 2017 • Q43 • ID: JAM_2017_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43 Consider the permutations $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 7 & 8 & 6 & 1 & 2 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 1 & 7 & 6 & 8 & 2 \end{pmatrix}$ in S_8 . The number of $\eta \in S_8$ such that $\eta^{-1}\sigma\eta = \tau$ is equal to _____

Question 36 (JAM 2017 • Q42 • ID: JAM_2017_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let G be a subgroup of $GL_2(\mathbb{R})$ generated by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$. Then the order of G is _____

Question 37 (JAM 2017 • Q35 • ID: JAM_2017_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35 Let G be a group of order 20 in which the conjugacy classes have sizes 1, 4, 5, 5, 5. Then which of the following is/are TRUE?

- (A) G contains a normal subgroup of order 5
- (B) G contains a non-normal subgroup of order 5
- (C) G contains a subgroup of order 10
- (D) G contains a normal subgroup of order 4

Question 38 (JAM 2017 • Q31 • ID: JAM_2017_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31 For $\alpha, \beta \in \mathbb{R}$, define the map $\varphi_{\alpha, \beta}: \mathbb{R} \rightarrow \mathbb{R}$ by $\varphi_{\alpha, \beta}(x) = \alpha x + \beta$. Let

$$G = \{\varphi_{\alpha, \beta} \mid (\alpha, \beta) \in \mathbb{R}^2\}$$

For $f, g \in G$, define $g \circ f \in G$ by $(g \circ f)(x) = g(f(x))$. Then which of the following statements is/are TRUE?

- (A) The binary operation $\circ$ is associative
- (B) The binary operation $\circ$ is commutative
- (C) For every $(\alpha, \beta) \in \mathbb{R}^2$, $\alpha \neq 0$ there exists $(a, b) \in \mathbb{R}^2$ such that $\varphi_{\alpha, \beta} \circ \varphi_{a, b} = \varphi_{1, 0}$
- (D) $(G, \circ)$ is a group

Question 39 (JAM 2017 • Q3 • ID: JAM_2017_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 The number of generators of the additive group $\mathbb{Z}_{36}$ is equal to

- (A) 6
- (B) 12
- (C) 18
- (D) 36

Question 40 (JAM 2016 • Q58 • ID: JAM_2016_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 Let H denote the group of all 2×2 invertible matrices over $\mathbb{Z}_5$ under usual matrix multiplication. Then the order of the matrix $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ in H is _____

Question 41 (JAM 2016 • Q50 • ID: JAM_2016_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 Let G be a cyclic group of order 12. Then the number of non-isomorphic subgroups of G is _____

Q. 51 – Q. 60 carry two marks each.

Question 42 (JAM 2016 • Q38 • ID: JAM_2016_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let G be a finite group and $o(G)$ denotes its order. Then which of the following statement(s) is(are) TRUE?

- (A) G is abelian if $o(G) = pq$ where p and q are distinct primes
- (B) G is abelian if every non identity element of G is of order 2
- (C) G is abelian if the quotient group $\frac{G}{Z(G)}$ is cyclic, where $Z(G)$ is the center of G
- (D) G is abelian if $o(G) = p^3$, where p is prime

Question 43 (JAM 2016 • Q24 • ID: JAM_2016_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 The sum of the series

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 + n - 2}$$

is

- (A) $\frac{1}{3} \ln 2 - \frac{5}{18}$ (B) $\frac{1}{3} \ln 2 - \frac{5}{6}$ (C) $\frac{2}{3} \ln 2 - \frac{5}{18}$ (D) $\frac{2}{3} \ln 2 - \frac{5}{6}$

Question 44 (JAM 2016 • Q14 • ID: JAM_2016_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 The value of the integral

$$\frac{(2n)!}{2^{2n} (n!)} \int_{-1}^1 (1-x^2)^n dx, \quad n \in \mathbb{N}$$

is

- (A) $\frac{2}{(2n+1)!}$ (B) $\frac{2n}{(2n+1)!}$
 (C) $\frac{2(n!)}{2n+1}$ (D) $\frac{(n+1)!}{2n+1}$

Question 45 (JAM 2016 • Q4 • ID: JAM_2016_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let σ be an element of the permutation group S_5 . Then the maximum possible order of σ is

(A) 5

(B) 6

(C) 10

(D) 15

Question 46 (JAM 2015 • Q52 • ID: JAM_2015_Q52)

NAT

+2 M

Neg: Nil (0)

Q.12 Suppose G is a cyclic group and $\sigma, \tau \in G$ are such that $\text{order}(\sigma) = 12$ and $\text{order}(\tau) = 21$. Then the order of the smallest group containing σ and τ is _____

Question 47 (JAM 2015 • Q47 • ID: JAM_2015_Q47)

NAT

+1 M

Neg: Nil (0)

Q.7

If $5^{2015} \equiv n \pmod{11}$ and $n \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then n is equal to _____

MA

14/16

Question 48 (JAM 2015 • Q43 • ID: JAM_2015_Q43)

NAT

+1 M

Neg: Nil (0)

Q.3

The number of distinct normal subgroups of S_3 is _____

Question 49 (JAM 2015 • Q32 • ID: JAM_2015_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.2

Which of the following statements is (are) true?

- (A) $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_6$
- (B) $\mathbb{Z}_3 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_9$
- (C) $\mathbb{Z}_4 \oplus \mathbb{Z}_6$ is isomorphic to $\mathbb{Z}_{24}$
- (D) $\mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5$ is isomorphic to $\mathbb{Z}_{30}$

MA

11/16

Question 50 (JAM 2015 • Q14 • ID: JAM_2015_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 Let S be the bounded surface of the cylinder $x^2 + y^2 = 1$ cut by the planes $z = 0$ and $z = 1 + x$. Then the value of the surface integral $\iint_S 3z^2 d\sigma$ is equal to

- (A) $\int_0^{2\pi} (1 + \cos \theta)^3 d\theta$
- (B) $\int_0^{2\pi} \sin \theta \cos \theta (1 + \cos \theta)^2 d\theta$
- (C) $\int_0^{2\pi} (1 + 2 \cos \theta)^3 d\theta$
- (D) $\int_0^{2\pi} \sin \theta \cos \theta (1 + 2 \cos \theta)^2 d\theta$

Question 51 (JAM 2015 • Q2 • ID: JAM_2015_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 Let $y(x) = u(x) \sin x + v(x) \cos x$ be a solution of the differential equation $y'' + y = \sec x$. Then $u(x)$ is

- (A) $\ln |\cos x| + C$
- (B) $-x + C$
- (C) $x + C$
- (D) $\ln |\sec x| + C$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Groups (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	9
2	NAT	+1	4
3	NAT	+1	90
4	MSQ	+2	A, B
5	MCQ	+2	A
6	MCQ	+2	C
7	MCQ	+2	C
8	MCQ	+2	C
9	MCQ	+2	A
10	MCQ	+1	B
11	NAT	+2	3 to 3
12	NAT	+1	3 to 3
13	MSQ	+2	C;D
14	MCQ	+2	C
15	MCQ	+2	D
16	MCQ	+2	B
17	MCQ	+1	B

Q. No.	Type	Marks	Official Key
18	MCQ	+1	D
19	NAT	+2	4 to 4
20	NAT	+2	31 to 31 or 46 to 46
21	NAT	+1	99 to 99
22	MSQ	+2	B, C
23	MCQ	+2	C
24	MCQ	+2	B
25	MCQ	+1	B
26	NAT	+1	0 to 0
27	NAT	+1	4 to 4
28	MSQ	+2	B;C;D
29	MSQ	+2	A;D
30	MCQ	+2	A
31	MCQ	+2	A
32	MCQ	+1	B
33	NAT	+2	29.9 to 30.1
34	NAT	+1	7.9 to 8.1

Q. No.	Type	Marks	Official Key
35	NAT	+1	−0.01 to +0.01
36	NAT	+1	5.9 to 6.1
37	MSQ	+2	A, C
38	MSQ	+2	A, C
39	MCQ	+1	B
40	NAT	+2	3.0 to 3.0
41	NAT	+1	6.0 to 6.0
42	MSQ	+2	B;C
43	MCQ	+2	C
44	MCQ	+2	C
45	MCQ	+1	B
46	NAT	+2	84
47	NAT	+1	1
48	NAT	+1	3
49	MSQ	+2	A;D
50	MCQ	+2	A
51	MCQ	+1	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (−1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.